

Siddharth Chatteraj

siddharthchatteraj@gmail.com | +1 (408) 747-9587 | Portfolio: <https://siddharthchatteraj.com/>

EDUCATION

University of California, Santa Barbara: 3.98 GPA

Santa Barbara, CA

B.S. Statistics and Data Science | B.A. Theater (Design) | Minor: Language and Speech Technologies September 2022 – March 2026

Senior Honors Project: Designing Embodied Wearable Technology to Enhance Kinesthetic Expression for Dance Performance

RESEARCH EXPERIENCE

Computational Psycholinguistics of Listening and Speaking Lab

Santa Barbara, CA

Undergraduate Research Assistant

January 2023 – Present

- Built models of simple dynamical systems of language change in Python by deriving and implementing formulas for statistical model analysis. Optimized parallel process and file writing runtimes from 120 minutes to 30 seconds. Currently writing a paper for submission to computational linguistics conferences and journals
- Developed an SQL tool to query word frequency and score data from 1,000 relational databases and determined best samples by calculating Jensen-Shannon divergence 1,000 times for 30 classifications of data using kernel density curves
- Conducted word-based and morph-based phonotactic scoring and analysis of 200,000 Spanish non-words using Python Pandas, SRILM, and Morfessor to generate a set of stimuli for an experiment examining Spanish proto-lexicons

CONFERENCE PRESENTATIONS AND EXHIBITIONS

- Ventura, Meredith and Delila Moseley (Lighting and Projection Design by Siddharth Chatteraj). 2025. Curio. *Dance Fest Skopje Opening Night* [performance]. Festival. Macedonian National Theatre, Skopje, Macedonia
 - Developed algorithmic projection design and lighting grounded in semiotic theory and 1920s–30s cabaret aesthetics to support a dance-theater investigation of gendered representation in media. Animated visual assets in After Effects
- Chatteraj, Siddharth, Shahil Patel, and Simon Todd. 2024. Deepening understanding of frequency effects in exemplar models of language change [poster]. Conference. *CAMP[7]: 7th Annual California Meeting on Psycholinguistics*. UC San Diego, CA
 - Authored a simulation-based study comparing decay and overwriting architectures in exemplar models using R. Visualized value and variance trajectories across 10,000+ iterations to reveal parameter-sensitive frequency effects
- Chatteraj, Siddharth. 2023. Realms of conception [installation]. *RARE: Realities Altered Realities Emerged*. Exhibition. Santa Barbara Center for Art, Science and Technology, Santa Barbara, CA
 - Designed an interactive audiovisual system using TouchDesigner, Python NLP sentiment analysis, and OSC. Users typed their thoughts to influence color based on emotional tone and adjusted visual scale using physical controls
- Cai, Grant, Henry Ma, Jessie Zhou, Siddharth Chatteraj, Stella Jia. 2023. Improving Online Legal Consultation: Strategies for Enhanced Client Assistance [presentation]. *DataFest Finalist Showcase*. Competition Hackathon. UC Los Angeles, CA. Finalist
 - Evaluated 850,000 American Bar Association Free Legal Answers records in 48 hours with Pandas/Natural Language Processing and utilized Python to design charts to propose accessibility improvements
- Nunez, Jesse, Madhav Viswesvaran, Ishaan Avinash, Calais Waring, Siddharth Chatteraj, Anna Lin. 2023. SlangSwitch [presentation and poster]. *Data Science UCSB Project Showcase*. Competition Conference. UC Santa Barbara, CA. 2nd Place
 - Developed a regional slang-to-formal text translator using Styleformer and web-scraped slang corpora. Applied Regex-based tagging and built an interactive demo to demonstrate sociolinguistic style transfer

TEACHING EXPERIENCE

UCSB Department of Computer Science

Santa Barbara, CA

Lead Undergraduate Learning Assistant (ULA)

April 2023 – Present

- Served as Lead ULA, mentoring newer ULAs and coordinating peer teaching strategies across seven quarters
- Designed LLM-integrated assessments for one of the first UCSB classes to incorporate generative AI into instruction, introducing students to data science through tools like prompt-based analysis, model-assisted coding, and language model evaluation
- Tutored weekly class sections of 30 beginner data science students in table operations, hypothesis testing, bootstrapping, statistical prediction techniques, linear regression, data visualization, clustering, and classification concepts
- Graded 100-200 weekly coding and analysis assignments and oral presentations and held weekly office hours to provide one-on-one support for conceptual understanding, homework, labs, and projects

Quarter

Course

Role

W24, S24, F24, W25, S25, F25, W26

CMPSC 5A: Introduction to Data Science 1 with Large Language Models

Lead ULA

F23

CMPSC 5A: Introduction to Data Science 1

ULA

S23

CMPSC 5B: Introduction to Data Science 2

ULA

WORK EXPERIENCE

Daily Nexus

Santa Barbara, CA

Data Editor and Games Section Founder

April 2023 – Present

- Oversaw the production of over 75 data-driven articles and mentored more than 100 student journalists on data storytelling, sourcing, and visualization, leading to multiple statewide awards: 1st Place for Best Multimedia Package (2025), 2nd Place for Best Interactive Graphic (2024), and 1st Place for Best Infographic (2023) from the California College Media Association
- Founded the Games section and launched its website, making the Nexus one of the first college newspapers with a dedicated games page. Designed interactive crosswords with JavaScript and taught workshops on data visualization, NLP, and web scraping
- Led management of The Labyrinth, a public dashboard visualizing 16 years of UCSB course grade data—over 40,000 classes—using D3.js. Grew weekly users from ~100 to over 1,000 through targeted marketing and feature expansion.

Assistant Data Editor

September 2022 – April 2023

- Self-taught Python to create multiple interactive charts and interviewed sources to compose articles about events across Santa Barbara. Obtained data from PRA requests and produced articles on strikes, historical crime trends, sports, and DEI

Amazon

Seattle, WA

Business Intelligence Engineer Intern

June 2025 – September 2025

- Saved 92 hours/week of manual analysis by building an AI agent that analyzes return cost per unit in 3–5 minutes via SQL and schema analysis and forecasts global trends with time-series/ML models, informing policy on return centers and backhaul carriers
- Developed two automated Cline AI-agent-creation workflows based on learnings from creating the cost per unit agent, enabling engineers to generate working agents in 5 minutes and cutting development time by 20% by starting at testing and refinement
- Investigated unreconciled EU returns by building SQL queries and automating ETL pipelines, analyzing backhaul carrier data to identify fulfillment inefficiencies and automate reporting
- Designed QuickSight dashboards tracking weekly, monthly, and quarterly transportation return metrics to support business planning and operational insights

Savie Health

Remote/Lompoc, CA

Revenue Manager

February 2025 – September 2025

- Developed forecasting algorithms to predict monthly Medi-Cal revenue with 95% accuracy and ensure timely payments
- Built Python SARIMA time series models to forecast patient attendance based on operating changes and revenue from Medi-Cal
- Designed reports on monthly enhanced care management statistics with Excel for submission to CenCal

Data Analyst

August 2024 – February 2025

- Designed and automated monthly reports to help the organization earn over \$65,000 in grants and make hiring decisions
- Rebranded organization's website using WordPress and assisted leadership with outreach communication

Data Analysis/Marketing Intern

July 2022 – August 2024

- Created promotional graphics and e-blasts in English and Spanish to advertise the startup nonprofit and helped secure over \$100,000 in funding and 1,500 monthly social media engagements through email marketing and data reports for grant requests

UCSB Department of Theater and Dance

Santa Barbara, CA

Lead Technician and Stage Manager

March 2023 – May 2025

- Led technical operations for 8 productions, managing a crew of 5-10 to run scenic/lighting transitions and fix equipment issues
- Stage managed 7 additional dance and theater productions, coordinating cues, rehearsals, and backstage flow for casts of ~20 actors and ~50 dancers while supervising 5-10 crew members and collaborating with 3-5 designers to execute the show

San Francisco District Attorney's Office

San Francisco, CA

San Francisco Sentencing Community Initiatives Data Fellow

June 2023 – September 2023

- Created a regenerable temporal and geospatial data memo for distribution to community members across the city using HTML, JavaScript, and Python Plotly to visualize crime trends in San Francisco supervisor and police districts
- Analyzed several databases of over 200,000 county crime records and visualized crime location trends using Geopandas, R, Python Plotly, and ArcGIS for grant requests. Wrote a 21-page guide on how to analyze and map geospatial data

DESIGN EXPERIENCE

Role	Production	Company	Date
Assistant Projection Designer	<i>TrollFest (Juneau)</i>	Ray Troll and the Ratfish Wranglers	05/2025
Technical Director/Lighting and Projection Designer/Stage Manager	<i>UCSB Dance Company (European Tour)</i>	UCSB Dance Company	04/2025
Lighting Designer	<i>Afterlight</i>	UCSB Division of Dance	04/2025
Projection Designer/Stage Manager	<i>Refractions: Voices in Light & Dark</i>	UCSB Dance Company	03/2025
Assistant Lighting Designer/Lead Technician	<i>Strange Birds</i>	LAUNCH PAD	02/2025
Lighting Designer/Stage Manager	<i>UCSB Dance Company Fundraiser</i>	UCSB Dance Company	02/2025
Projection Designer	<i>The Threepenny Opera</i>	UCSB Division of Theater	11/2024
Projection Designer/Lead Technician	<i>Kinetic Lab</i>	UCSB Division of Dance	06/2024
Assistant Projection Designer	<i>Indecent</i>	UCSB Division of Theater	05/2024